

25-10-29_update

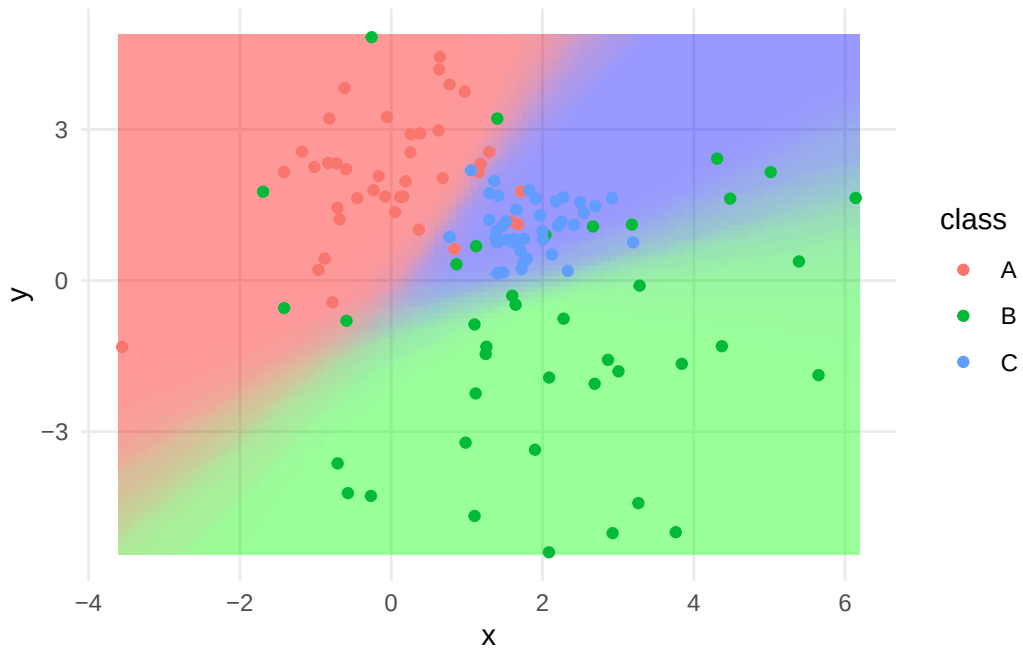
Update to make_template creating a file and opening into new window

Create a wrapper function in ggclassify for blending color

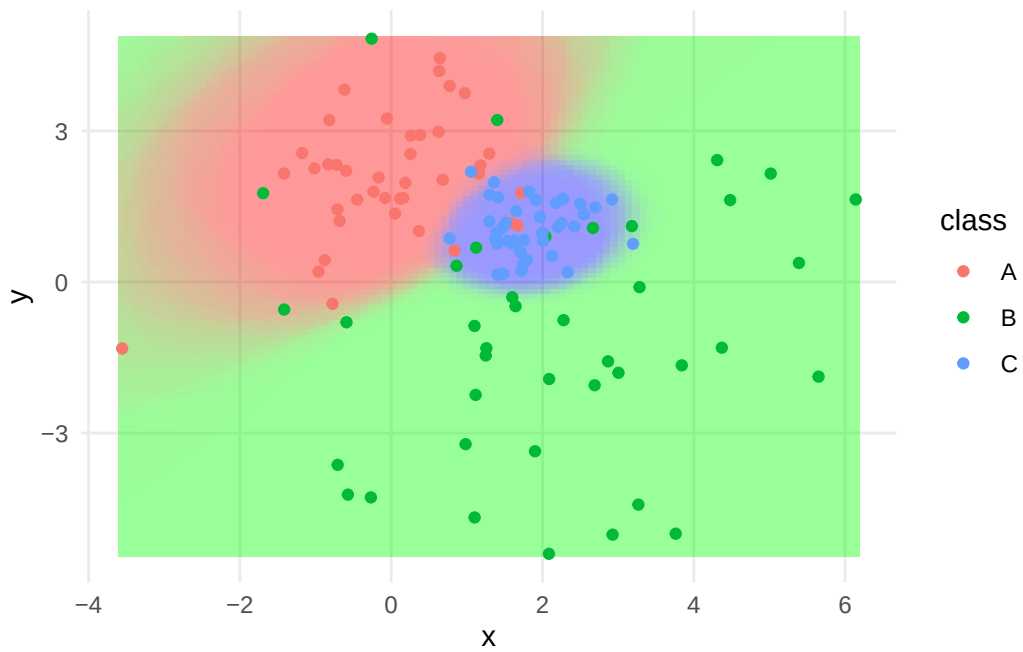
Boundary bootwrapping function

uses rgb to return the proportion of each values that belongs to each class on the mesh

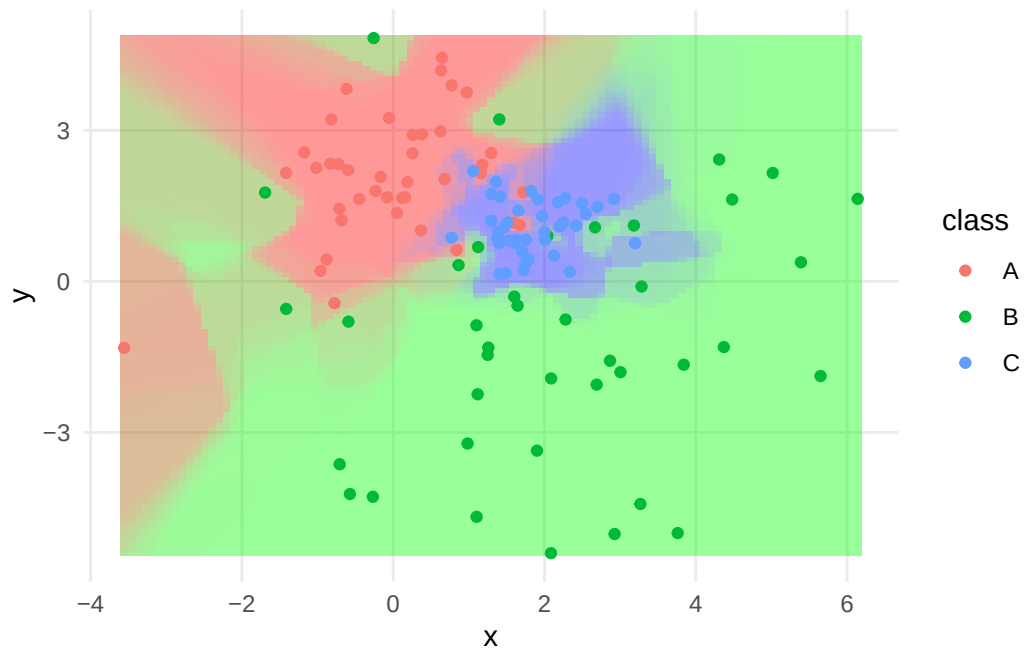
```
simpleplot <- ggplot(df, aes(x, y, fill = class))+  
  geom_point_fill()  
  
classify_boot_blend(df, simpleplot, n = 100, res = 101, method = "lda", blend = "tricolor")
```



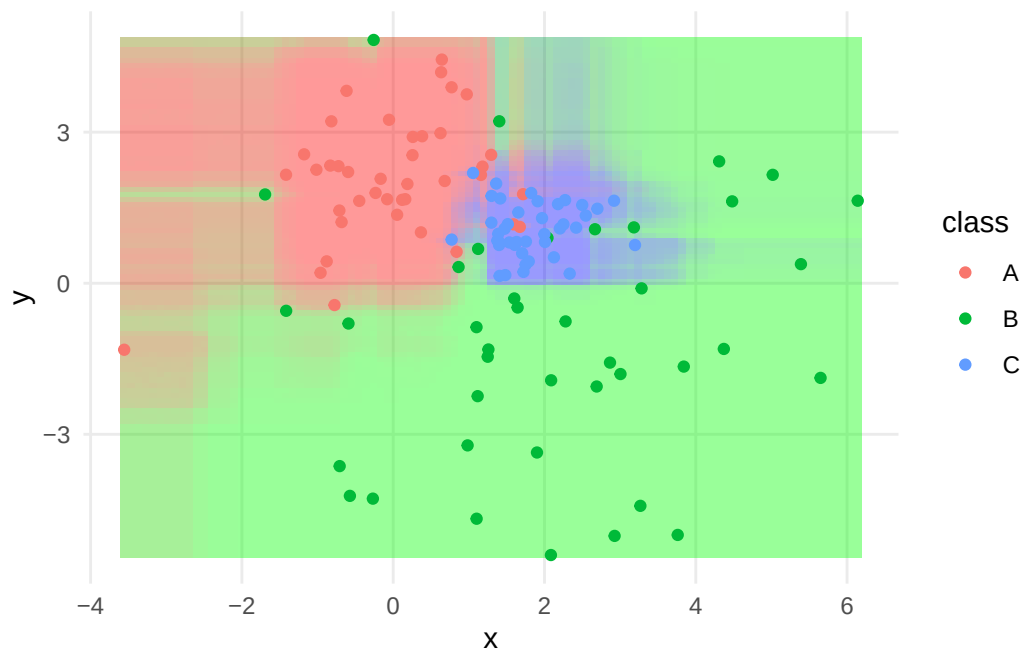
```
classify_boot_blend(df, simpleplot, n = 100, res = 101, method = "qda", blend = "tricolor")
```



```
classify_boot_blend(df, simpleplot, n = 100, res = 101, method = "knn", blend = "tricolor")
```

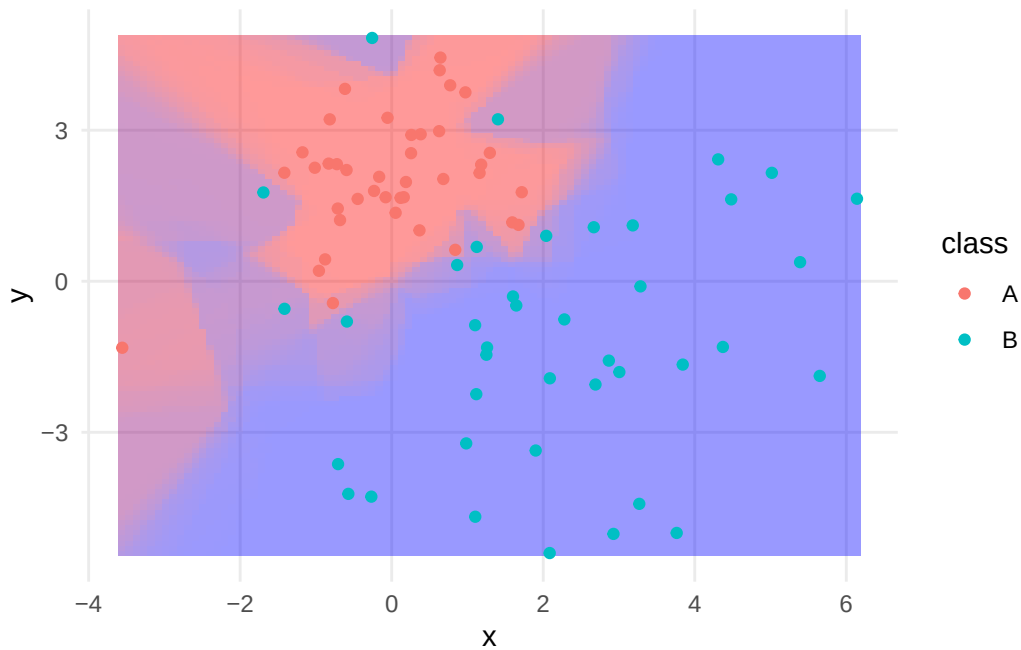


```
classify_boot_blend(df, simpleplot, n = 100, res = 101, method = "randomForest", blend = "tricolor")
```



```
simpleplot_2 <- ggplot(df |> filter(class %in% c("A","B")),aes(x,y,fill = class))+
  geom_point(aes(col = class))

df |> filter(class %in% c("A","B")) |>
  classify_boot_blend(simpleplot_2, n = 100, res = 101,
    method = "knn", blend = "bicolor")
```



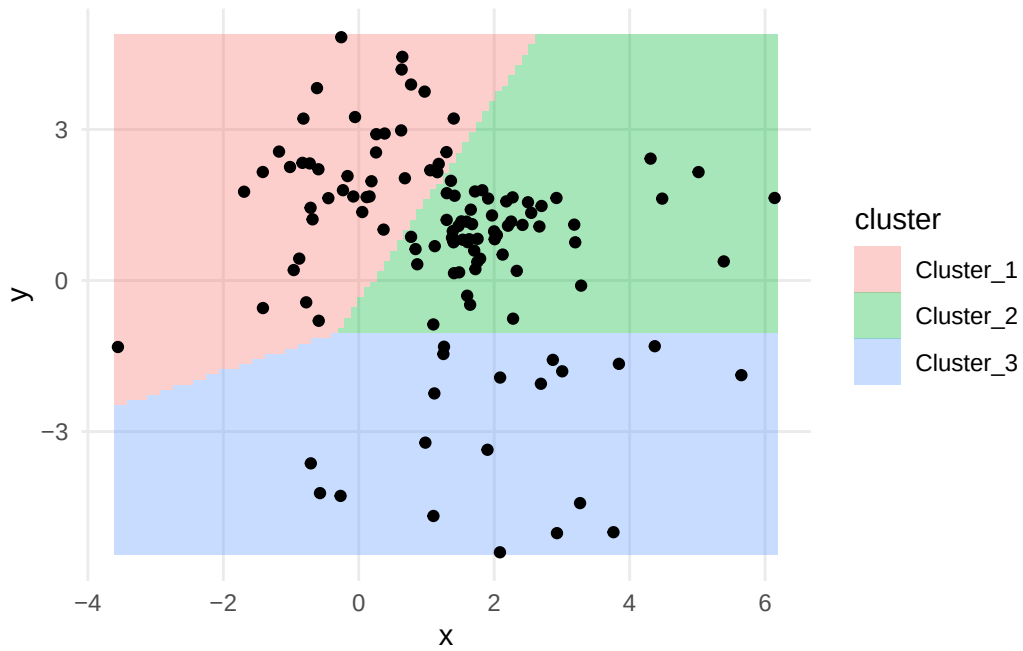
Clustering Geoms

Geom_Cluster

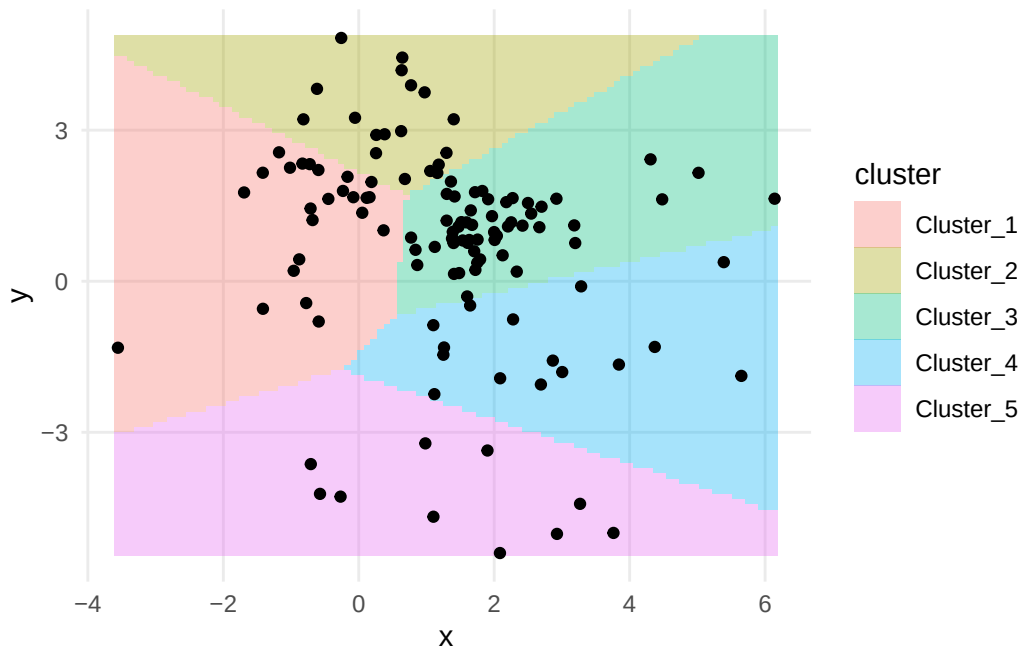
```
library(tidyclust)

kmeans_spec <- k_means(num_clusters = 3) |> set_engine("stats")
kmeans_spec_5 <- k_means(num_clusters = 5) |> set_engine("stats")

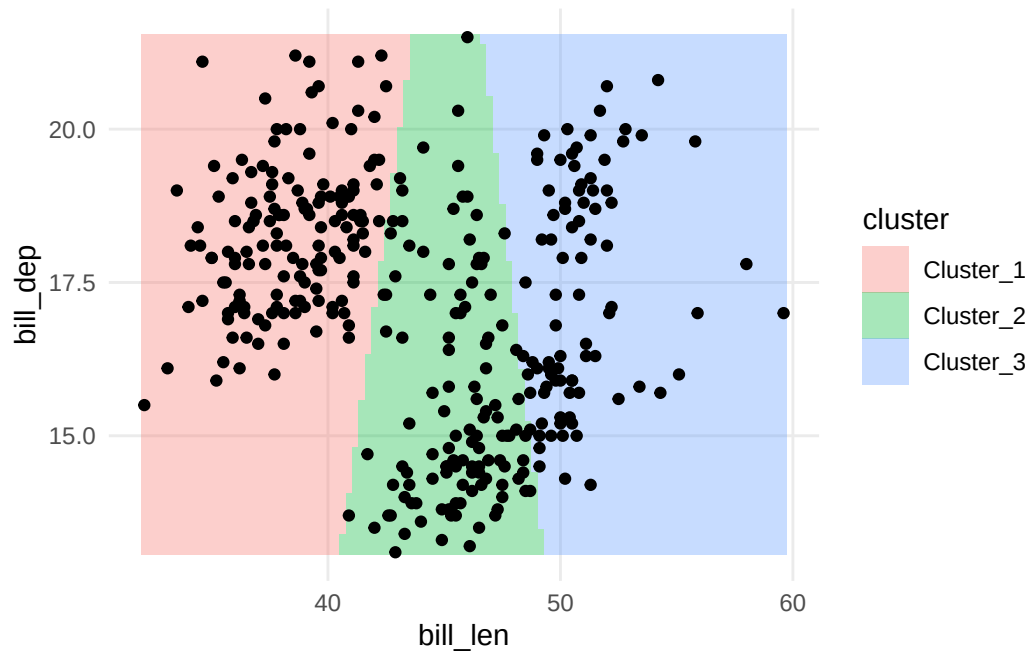
ggplot(df, aes(x,y)) +
  geom_cluster(method = kmeans_spec) +
  geom_point()
```



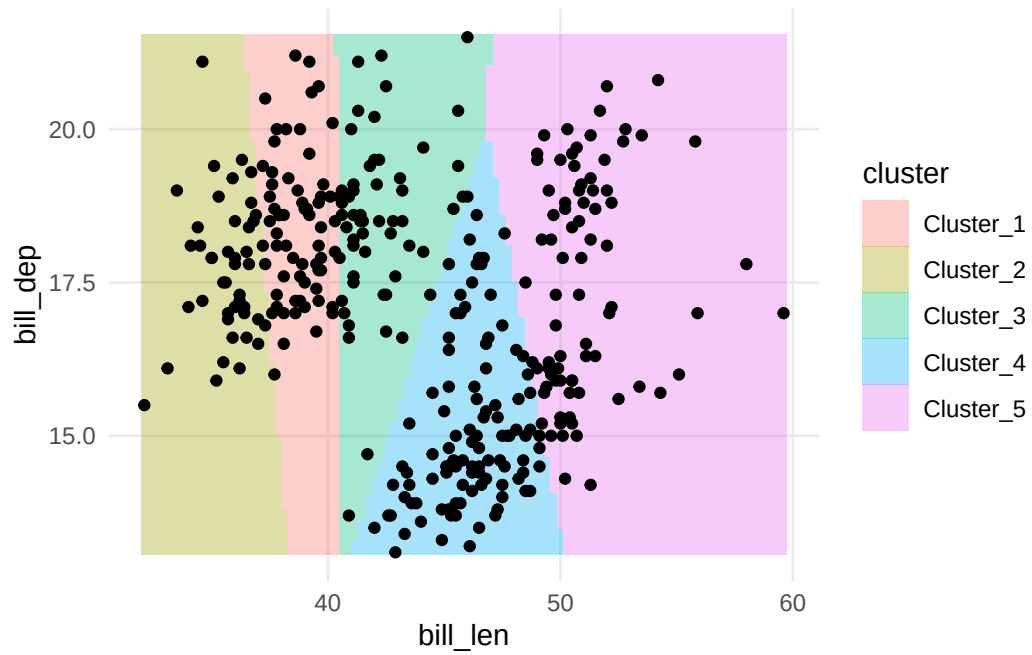
```
ggplot(df, aes(x,y)) +  
  geom_cluster(method = kmeans_spec_5) +  
  geom_point()
```



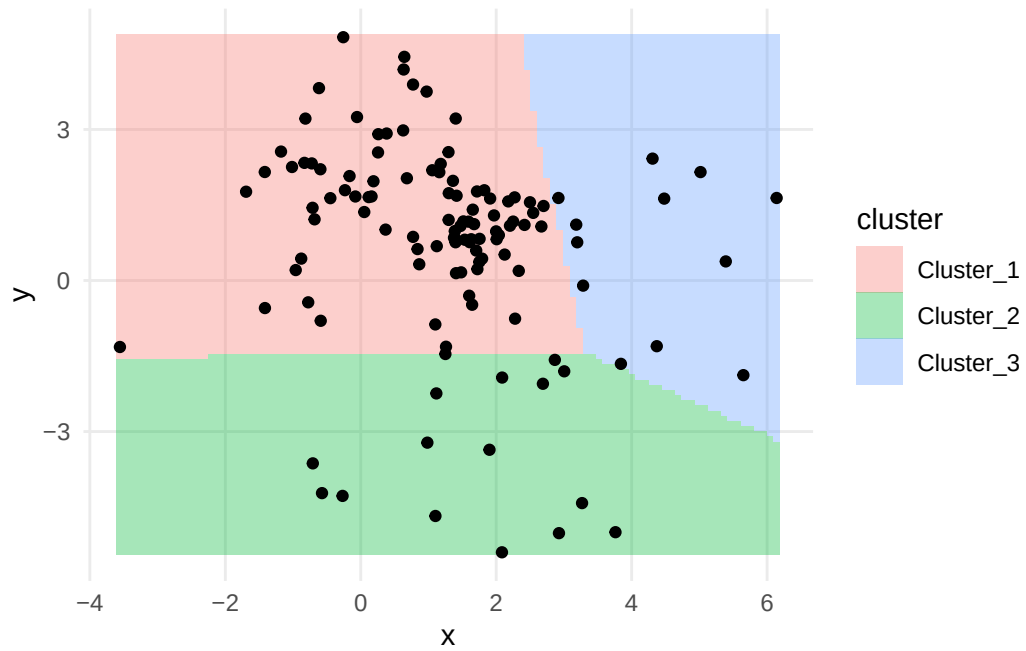
```
ggplot(penguins, aes(bill_len, bill_dep)) +  
  geom_cluster(method = kmeans_spec) +  
  geom_point()
```



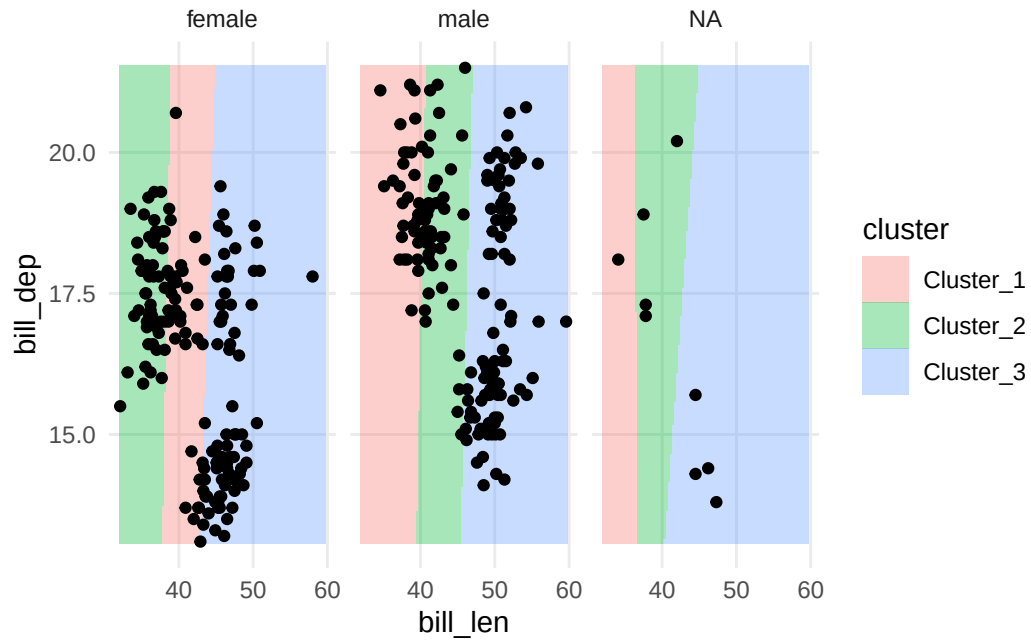
```
ggplot(penguins, aes(bill_len, bill_dep)) +  
  geom_cluster(method = kmeans_spec_5) +  
  geom_point()
```



```
hc_spec <- hier_clust(  
  num_clusters = 3,  
  linkage_method = "average"  
)  
  
ggplot(df, aes(x, y)) +  
  geom_cluster(method = hc_spec) +  
  geom_point()
```

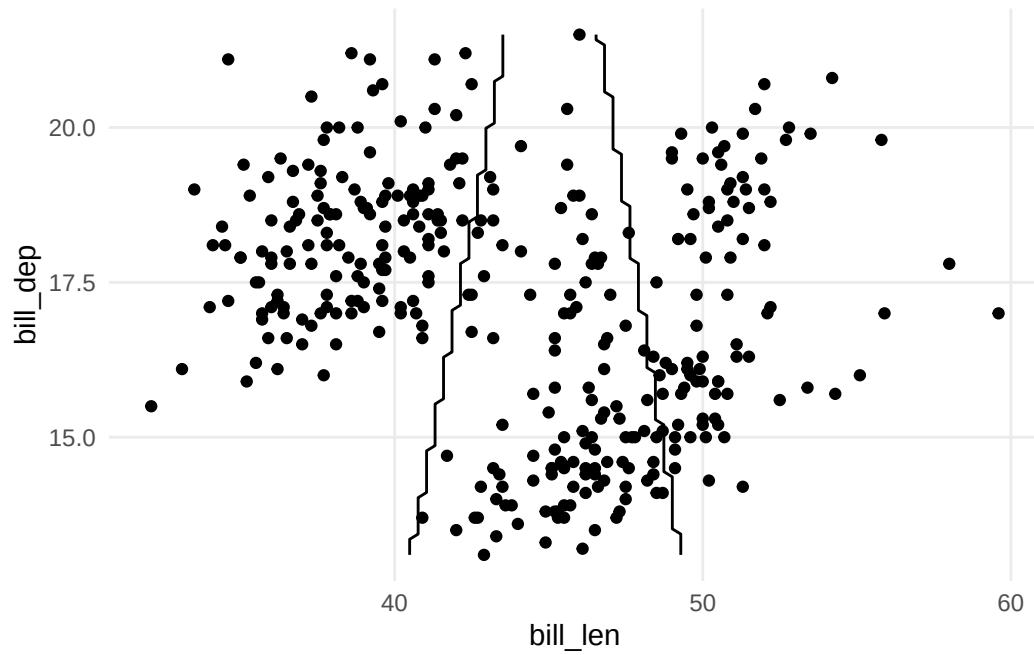


```
ggplot(penguins, aes(bill_len, bill_dep)) +  
  geom_cluster(method = kmeans_spec) +  
  geom_point() +  
  facet_wrap(~sex)
```

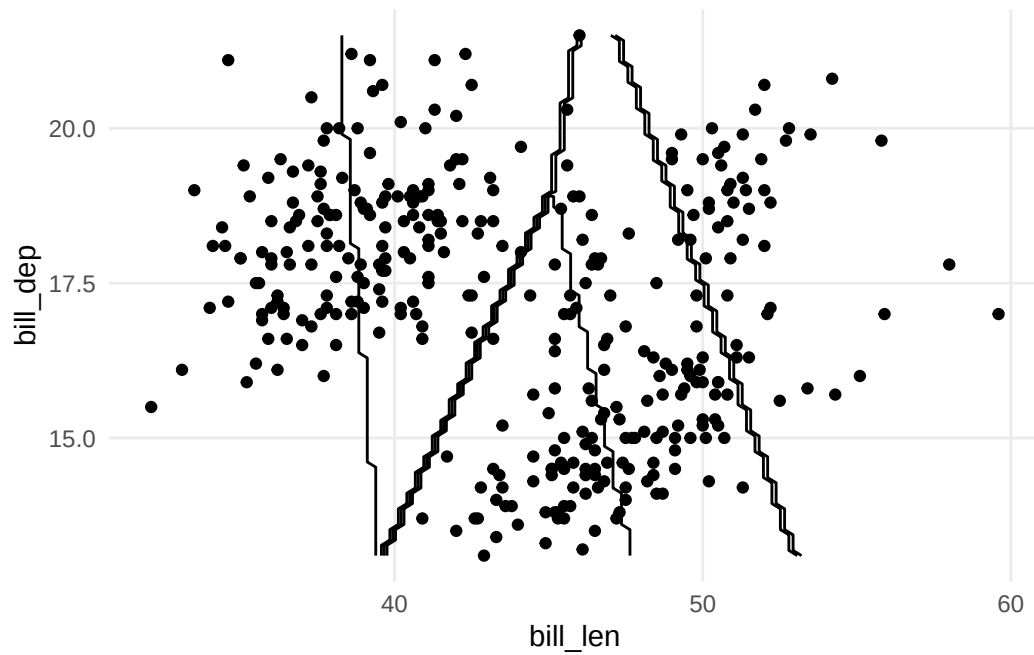



Geom_Cluster_Boundary

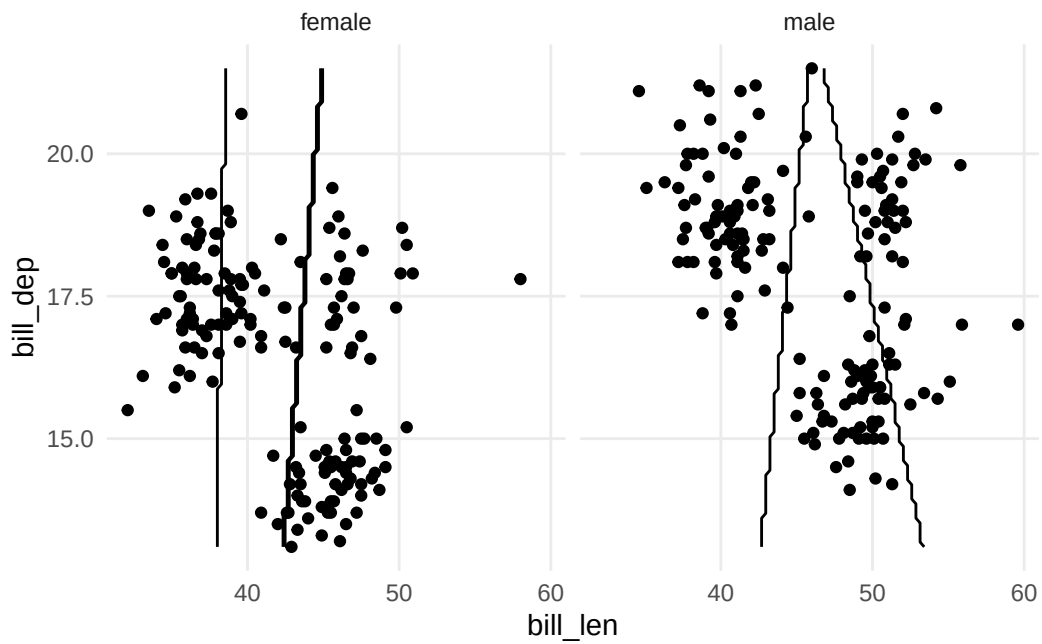
```
ggplot(penguins, aes(bill_len, bill_dep)) +  
  geom_cluster_boundary(method = kmeans_spec) +  
  geom_point()
```



```
ggplot(penguins, aes(bill_len, bill_dep)) +  
  geom_cluster_boundary(method = kmeans_spec_5) +  
  geom_point()
```



```
ggplot(penguins |> remove_missing(), aes(bill_len, bill_dep)) +
  geom_cluster_boundary(method = kmeans_spec) +
  geom_point() +
  facet_wrap(~sex)
```



```
ggplot(penguins |> remove_missing(), aes(bill_len, bill_dep)) +
  geom_cluster_boundary(method = kmeans_spec, aes(lty = sex)) +
  geom_point()
```

